Testing in generalized partly linear models: A robust approach *

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Abstract

In this paper, we introduce a family of robust statistics which allow to decide between a parametric model and a semiparametric one. More precisely, under a generalized partially linear model, i.e., when the observations satisfy $y_i|\left(\mathbf{x}_i, t_i\right) \sim F\left(\cdot, \mu_i\right)$ with $\mu_i = H\left(\eta(t_i) + \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}\right)$ and H a known link function, we want to test $H_0 : \eta(t) = \alpha + \gamma t$ against $H_1 : \eta$ is a nonlinear smooth function. A general approach which includes a robust version of the deviance and a robustified quasi–likelihood is considered. The asymptotic behavior of the test statistic under the null hypothesis is obtained.

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1 Introduction

Semiparametric models contain both a parametric and a nonparametric component. Sometimes the nonparametric component plays the role of a nuisance parameter. A lot of research has been done on estimators of the parametric component in a general framework, aiming to obtain asymptotically efficient estimators. The aim of this paper is to consider semiparametric versions of the generalized linear models where the response y is to be predicted by covariates $(\mathbf{x}, t)$, where $\mathbf{x} \in \mathbb{R}^p$ and $t \in \mathcal{T} \subset \mathbb{R}$ with $\mathcal{T}$ a compact set. Without loss of generality we will assume that $\mathcal{T} = [0, 1]$. It will also be assumed that the conditional distribution of $y|(\mathbf{x}, t)$ belongs to the canonical exponential family $\exp\left[y\theta(\mathbf{x}, t) - B\left(\theta(\mathbf{x}, t)\right) + C(y)\right]$, for known functions B and C. Then, $\mu\left(\mathbf{x}, t\right) = \mathbb{E}\left(y|(\mathbf{x}, t)\right) = B'\left(\theta(\mathbf{x}, t)\right)$, with B' as the derivative of B. In generalized linear models (McCullagh and Nelder, 1989), which is a popular technique for modelling a wide variety of data, it is often assumed that the mean is modelled linearly through a known link function, g, i.e., $g(\mu\left(\mathbf{x}, t\right)) = \beta_0 + \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta} + \alpha t$. For instance, an ordinary logistic regression model assumes that the observations $(y_i, \mathbf{x}_i, t_i)$ are such that the responses are independent binomial variables $y_i|(\mathbf{x}_i, t_i) \sim Bi(1, p_i)$ whose success probabilities depend on the explanatory variables through the relation $g(p_i) = \beta_0 + \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + \alpha t_i$, with $g(u) = \log\left(u/(1-u)\right)$.

In many situations, the linear model is insufficient to explain the relationship between the response variable and its associated covariates. A natural generalization, which suffers from the *curse of dimensionality*, is to model the mean nonparametrically in the covariates. An alternative strategy is to allow most predictors to be modeled linearly while one or a small number of predictors enter the model nonparametrically. This is the approach we will follow, so that the relationship will be given by the semiparametric generalized partially linear model

$$\mu\left(\mathbf{x}, t\right) = H\left(\eta(t) + \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}\right) \tag{1}$$

where $H = g^{-1}$ is a known link function, $\boldsymbol{\beta} \in \mathbb{R}^p$ is an unknown parameter and η is an unknown smooth function.

In the context of hypothesis testing, when $H(t) = t$, Gao (1997) established a large sample theory for testing $H_0 : \boldsymbol{\beta} = 0$ and, in addition to this, Härdle *et al.* (2000) tested $H_{0,\eta} : \eta = \eta_0$ too. Recently, González–Manteiga and Aneiros–Pérez (2003) studied the case of dependent errors. For generalized partly linear models, Härdle *et al.* (1998) considered a test statistic to decide between a linear and a semiparametric model. Their proposal is based on the estimation procedure considered by Severini and Staniswalis (1994) modified to deal with the smoothed and unsmoothed likelihoods. A comparative study of different procedures was performed by Müller (2001) while a different approach was considered in Rodríguez Campos *et al.* (1998).

As it is well known, such estimates fail to deal with outlying observations and so does the test statistic. In a semiparametric setting, outliers can have a devastating effect, since the extreme points can easily affect the scale and the shape of the function estimate of η, leading to possibly wrong conclusions. In particular, as mentioned Hampel's comment on Stone (1977) paper *"If we believe in a smooth model without spikes,..., some robustification is possible. In this situation, a clear outlier will not be attributed to some sudden change in the true model, but to a gross error, and hence it may be deleted or otherwise made harmless"*. Therefore, in this context robust procedures need to be developped to avoid wrong conclusions on the hypothesis to be tested (see Bianco *et al.* (2006) for a discussion).

Robust procedures for generalized linear models have been considered among others by Stefanski *et al.* (1986), Künsch *et al.* (1989), Bianco and Yohai (1995), Cantoni and Ronchetti (2001), Croux and Haesbroeck (2002) and Bianco *et al.* (2005). The basic ideas from robust smoothing and from robust regression estimation have been adapted to deal with the case of independent observations following a partly linear regression model with $H(t) = t$; we refer to Gao and Shi (1997), He *et al.* (2002) and Bianco and Boente (2004). Moreover, robust tests for a given alternative, under a partly linear regression model were studied in Bianco *et al.* (2006). Besides, a robust approach for testing the parametric form of a regression function versus an omnibus alternative, based on the centered asymptotic rank transformation, was considered by Wang and Qu (2007) when $H(t) = t$ and $\boldsymbol{\beta} = 0$, i.e., under the nonparametric model $y_i = \eta_0(t_i) + \epsilon_i$.

Under a generalized partially linear model (1), Boente *et al.* (2006) introduced a general profile–based two–step robust procedure to estimate the parameter $\boldsymbol{\beta}$ and the function η while Boente and Rodriguez (2010) (see also, Rodriguez, 2008) developed a three–step method to improve the computational time of the previous one. Beyond the importance of developing robust estimators in more general settings, the work on testing also deserves attention. An uptodate review of robust hypothesis testing results can be found in He (2002). The aim of this paper is to propose

a class of tests for the nonparametric component based on the three–step robust procedure proposed by Boente and Rodriguez (2010).

The paper is organized as follows. In Section 2, we remind the definition of the general profile–based two–step estimators as well as the three–step robust estimates and their asymptotic properties. In Section 3, we present a robust alternative to test hypothesis concerning the nonparametric component η. Their asymptotic behavior is studied in Section 4 while a boostrap procedure is discussed in Section 5. In Section 6, we present the results of a Monte Carlo study. Finally, in Section **??** we give some final conclusions. Proofs are relegated to the Appendix.

2 Preliminaries: The estimation procedure

As mentioned in the Introduction, Boente *et al.* (2006) introduced a highly robust procedure under model (1) while Boente and Rodriguez (2010) introduced a local approach to improve the computational time. Let $(y_i, \mathbf{x}_i, t_i)$ be independent observations such that $y_i|\left(\mathbf{x}_i, t_i\right) \sim F\left(\cdot, \mu_i\right)$ with $\mu_i = H\left(\eta(t_i) + \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}\right)$ and $\mathrm{VAR}\left(y_i|(\mathbf{x}_i, t_i)\right) = V\left(\mu_i\right)$. Let $\eta_0(t)$ and $\boldsymbol{\beta}_0$ denote the true parameter values, and $\mathbb{E}_0$ the expected value under the true model, so that $\mathbb{E}_0(y_1|(\mathbf{x}_1, t_1)) = H\left(\eta_0(t_1) + \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0\right)$.

As in Robinson (1988), we will assume that the vector $\mathbf{1}_n$ is not in the space spanned by the column vectors of $(\mathbf{x}_1, \cdots, \mathbf{x}_n)^{\mathrm{T}}$, that is, we do not allow $\boldsymbol{\beta}_0$ to include an intercept so that the model is identifiable, i.e., if $\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_1 + \eta_1(t_i) = \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_2 + \eta_2(t_i)$ for $1 \le i \le n$, then, $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$ and $\eta_1 = \eta_2$. Due to the generality of the semiparametric model (1), identifiability implies that only "slope"coefficients can be estimated. Moreover, as in the partly linear regression model, the hypothesis required to ensure asymptotic normality entail that we avoid any linear combination of the components of $\mathbf{x}$ from being a function of t. As Robinson (1988) noticed, this rules out the situation of an unknown regression function depending only on t, where $\boldsymbol{\beta}_0^{\mathrm{T}}\mathbf{x}$ represents the Taylor's expansion of order p and η taking care of the remainder term, since these models are more nonparametric than semiparametric.

Let $w_1 : \mathbb{R}^p \to \mathbb{R}$ be a weight function to control leverage points on the carriers $\mathbf{x}$, $\rho : \mathbb{R}^2 \to \mathbb{R}$ a loss function and $K : \mathbb{R} \to \mathbb{R}$ a kernel function. Define $S(\boldsymbol{\beta}, a, t) = \mathbb{E}_0\left[\rho\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta} + a\right) w_1(\mathbf{x}_1)|t_1 = t\right]$ and $S_n(\boldsymbol{\beta}, a, t) = \sum_{i=1}^n W_i(t)\rho\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + a\right) w_1(\mathbf{x}_i)$ where $W_i(t)$ are the kernel weights on t_i, i.e.

$$W_i(t) = \left(\sum_{j=1}^n K((t - t_j)/h_n)\right)^{-1} K((t - t_i)/h_n)\ .$$

It is worth noticing that the function $S(\boldsymbol{\beta}, a, t)$ is a continuous function of $(a, \boldsymbol{\beta}, t)$ if the distribution function of $(y_1, \mathbf{x}_1)|t_1 = t$ is continuous with respect to t. Moreover, let $F(\boldsymbol{\beta}, v) = \mathbb{E}_0\left[\rho\left(y, \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta} + v(t)\right) w_2(\mathbf{x})\right]$ and $F_n(\boldsymbol{\beta}, v) = (1/n)\sum_{i=1}^n \rho\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + v(t_i)\right) w_2(\mathbf{x}_i)$, where $w_2(\cdot)$ plays the same role (and can be taken to be the same) as $w_1(\cdot)$.

Note that $S_n(\boldsymbol{\beta}, a, t)$ is an estimate of $S(\boldsymbol{\beta}, a, t)$. Fisher–consistency states that $S(\eta_0(t), \boldsymbol{\beta}_0, t) = \min_a S(a, \boldsymbol{\beta}_0, t)$.

Boente *et al.* (2006) and Boente and Rodriguez (2010) considered two classes of loss functions $\rho(y, a)$. The first one aims to bound the deviances, while the second one introduced by Cantoni and Ronchetti (2001) bounds the Pearson residuals. We refer to these authors for a discussion on their choice, where also conditions to ensure Fisher–consistency of the resulting estimators are stated. For the sake of completeness, we remind the definition of the estimators introduced in Boente *et al.* (2006) and Boente and Rodriguez (2010).

Following the ideas of Severini and Staniswalis (1994), Boente *et al.* (2006) defined the function $\eta_{\boldsymbol{\beta}}(t)$ as the minimizer of $S(\boldsymbol{\beta}, a, t)$ that will be estimated by the minimizer $\widehat{\eta}_{\boldsymbol{\beta}}(t)$ of $S_n(\boldsymbol{\beta}, a, t)$. A two–step robust quasi–likelihood proposal was then defined as follows

- **Step BHZ1**: For each value of t and $\boldsymbol{\beta}$, let $\widehat{\eta}_{\boldsymbol{\beta}}(t) = \mathrm{argmin}_{a\in\mathbb{R}}\, S_n(\boldsymbol{\beta}, a, t)$.
- **Step BHZ2**: Define the estimate of $\boldsymbol{\beta}_0$ as $\widehat{\boldsymbol{\beta}}_{\text{BHZ}} = \mathrm{argmin}_{\boldsymbol{\beta}\in\mathbb{R}^p}\, F_n(\boldsymbol{\beta}, \widehat{\eta}_{\boldsymbol{\beta}})$ and the estimate of $\eta_0(t)$ as $\widehat{\eta}_{\text{BHZ}}(t) = \widehat{\eta}_{\widehat{\boldsymbol{\beta}}_{\text{BHZ}}}(t)$.

If the function $\rho(y,u)$ is continuously differentiable and we denote $\Psi(y,u) = \partial\rho(y,u)/\partial u$, the functional $\eta_{\boldsymbol{\beta}}(t)$ and the estimates $\widehat{\eta}_{\boldsymbol{\beta}}(t)$ will be a solution of the differentiated equations, i.e., they will be a solution of $S^{(1)}(\boldsymbol{\beta},a,t) = 0$ and $S_n^{(0,1)}(\boldsymbol{\beta},a,t) = 0$ respectively, where $S^{(1)}(\boldsymbol{\beta},a,\tau) = \mathbb{E}_0\left(\Psi\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a\right)w_1(\mathbf{x})|t=\tau\right)$ and $S_n^{(0,1)}(\boldsymbol{\beta},a,t) = \sum_{i=1}^n W_{0,i}(t)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}+a\right)w_1(\mathbf{x}_i)$ with

$$W_{0,i}(t) = \frac{1}{nh}K\left(\frac{t_i-t}{h_n}\right) . \tag{2}$$

Besides, $\widehat{\boldsymbol{\beta}}_{\text{BHZ}}$ is also the solution of a differentiated equation (see, Boente *et al.*, 2006 for details). These estimates turn out to be consistent and the estimates of $\boldsymbol{\beta}$ have a root-n convergence rate.

Since they are based on a profile–likelihood approach, these estimates may be computationally expensive, specially for high dimensional covariates $\mathbf{x}$. For that reason, Boente and Rodriguez (2010) introduced a new family of estimators of $\boldsymbol{\beta}$ and η as follows

- **Step R1.** For each value of t, let $(\widetilde{\beta}(t),\widehat{\eta}(t)) = \operatorname{argmin}_{\boldsymbol{\beta}\in\mathbb{R}^p, a\in\mathbb{R}} S_n(\boldsymbol{\beta},a,t)$.
- **Step R2.** Define the initial estimates of $\eta_0(t)$ as $\widehat{\eta}(t)$ and the estimate of $\boldsymbol{\beta}_0$, as $\widehat{\boldsymbol{\beta}}_{\text{R}} = \operatorname*{argmin}_{\boldsymbol{\beta}\in\mathbb{R}^p} F_n(\boldsymbol{\beta},\widehat{\eta})$.
- **Step R3.** Define the final estimate of η_0 as follows. For each value of t, let $\widehat{\eta}_{\text{R}}(t) = \operatorname{argmin}_{a\in\mathbb{R}} S_n(\widehat{\boldsymbol{\beta}}_{\text{R}},a,t)$.

Note that $\widehat{\eta}_{\text{R}}$ is simply $\widehat{\eta}_{\widehat{\boldsymbol{\beta}}_{\text{R}}}$. As in Boente *et al.* (2006), the estimators $\widehat{\boldsymbol{\beta}}_{\text{R}}$ and $\widehat{\eta}_{\text{R}}$ of $\boldsymbol{\beta}$ and η turn out to be consistent. Besides, the estimators $\widehat{\boldsymbol{\beta}}_{\text{R}}$ of $\boldsymbol{\beta}$ have also a root-n rate of convergence (see Boente and Rodriguez, 2010, for details).

3 Test statistics

A robust test statistic to test $H_0 : \eta \in \{\alpha+\gamma\, t,\ \alpha\in\mathbb{R},\gamma\in\mathbb{R}\}$ can be defined by comparing the semiparametric estimates $\widehat{\eta}_{\text{BHZ}}$ or $\widehat{\eta}_{\text{R}}$ defined in Section 2 with the estimators obtained under a parametric model. We will give an approach that robustifies the test statistic defined in Härdle *et al.*(1998).

Denote $\widehat{\boldsymbol{\beta}}$ a robust root$-n$ estimator of $\boldsymbol{\beta}$ and $\widehat{\eta}(t) = \widehat{\eta}_{\widehat{\boldsymbol{\beta}}}(t)$ the estimates of $\eta(t)$ solution of $\widehat{\eta}_{\widehat{\boldsymbol{\beta}}}(t) = \operatorname{argmin}_{a\in\mathbb{R}} S_n(\widehat{\boldsymbol{\beta}},a,t)$. Denote by $L(\boldsymbol{\beta},\alpha,\gamma) = \mathbb{E}_0\left[\rho\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+\alpha+\gamma t\right)w_2(\mathbf{x})\right]$ and

$$L_n(\boldsymbol{\beta},\alpha,\gamma) = \frac{1}{n}\sum_{i=1}^n \rho\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}+\alpha+\gamma t_i\right)w_2(\mathbf{x}_i)\,,$$

which correspond to the robustified objective functions under a generalized linear regression model.

Define the robust estimates under the linear model as the minimizer of L_n, i.e.,

$$(\widehat{\boldsymbol{\beta}}_{\text{H}_0},\widehat{\alpha}_{\text{H}_0},\widehat{\gamma}_{\text{H}_0}) = \operatorname*{argmin}_{\boldsymbol{\beta}\in\mathbb{R}^p,\alpha\in\mathbb{R},\gamma\in\mathbb{R}} L_n(\boldsymbol{\beta},\alpha,\gamma) \tag{3}$$

and denote $\widetilde{\Psi}(\mu,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a) = \mathbb{E}\left(\Psi\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a\right)|(\mathbf{x},t)\right)$ the conditional expectation when $y|(\mathbf{x},t)\sim F(\cdot,\mu)$. Let $\widetilde{y}_i = H\left(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\text{H}_0}+\widehat{\alpha}_{\text{H}_0}+\widehat{\gamma}_{\text{H}_0}t_i\right)$ stand for the pseudo–observations.

The function $\widehat{\eta}_{\text{H}_0}$ is defined as follows. Since the pseudo–observations will not have outliers, in the sense of large Pearson residuals, but only leverage points could appear, it is quite natural to define $\widehat{\eta}_{\text{H}_0}(t)$ as the value solving $\sum_{i=1}^n W_i(t)\widetilde{\Psi}(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\text{H}_0}+a)w_1(\mathbf{x}_i) = 0$, or equivalently as the value

$$\widehat{\eta}_{\text{H}_0}(t) = \operatorname*{argmin}_{a\in\mathbb{R}} \widetilde{S}_n(\widehat{\boldsymbol{\beta}}_{\text{H}_0},a,t)$$

where $\widetilde{S}_n(\boldsymbol{\beta}, a, t) = \sum_{i=1}^n W_i(t)\widetilde{\rho}\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + a\right) w_1(\mathbf{x}_i)$, with $\left(\partial\widetilde{\rho}(\mu, a)\right)/\partial a = \widetilde{\Psi}(\mu, a)$. Note that under mild conditions $\widetilde{\rho}(\mu, a) = \mathbb{E}\left(\rho\left(y, a\right)|(\mathbf{x}, t)\right)$ where the conditional expectation is taken when $y|(\mathbf{x}, t) \sim F(\cdot, \mu)$

Then, the test statistic is defined using a robust goodness–of–fit measure, based on the quasi–likelihood function

$$\mathcal{T}_1 = -2\sum_{i=1}^n Q\left(H\left(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}} + \widehat{\eta}(t_i)\right), H\left(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} + \widehat{\eta}_{\mathrm{H}_0}(t_i)\right)\right) w_2(\mathbf{x}_i)w(t_i)$$

where $Q(y, \mu) = \int_\mu^y (s - y)V^{-1}(s)\, ds$ is the quasi–likelihood. We are not considering a robust version of the quasi–likelihood, since only leverage points can appear on the expression on which the quasi–likelihood is to be computed and the weight function $w_2(\mathbf{x}_i)$ bounds their effect. We have also included a weight function $w(t)$ to avoid boundary effects. The function w has a compact support $\mathcal{T}_0 \subset \mathcal{T} = [0, 1]$, in particular we have that for n large enough $I_{[h_n, 1-h_n]}(t) \geq w(t)$.

4 Asymptotic behavior

For the sake of simplicity, we denote $\rho_n = h_n^2 + (n\, h_n)^{-\frac{1}{2}}$, $\chi(y, a) = \partial\Psi\left(y, a\right)/\partial a$, $\chi_1(y, a) = \partial^2\Psi\left(y, a\right)/\partial a^2$, $\widehat{\upsilon}(\boldsymbol{\beta}, t) = \widehat{\eta}_{\boldsymbol{\beta}}(t) - \eta_{\boldsymbol{\beta}}(t)$, $\widehat{\upsilon}_0(t) = \widehat{\upsilon}(\boldsymbol{\beta}_0, t)$, $\widehat{\upsilon}_j(\boldsymbol{\beta}, t) = \partial\widehat{\upsilon}(\boldsymbol{\beta}, t)/\partial\beta_j$ and $\widehat{\upsilon}_{j,0}(t) = \widehat{\upsilon}_j(\boldsymbol{\beta}_0, t)$.

We will need the following set of assumptions

A1. The density f of t_1 is bounded on $\mathcal{T}$, twice continuously differentiable in the interior of $\mathcal{T}$ with bounded derivatives

A2. $\inf_{t\in[0,1]} |f(t)| > 0$

A3. η_0 is twice continuously differentiable in the interior of $\mathcal{T}$ with bounded derivatives on $\mathcal{T}$.

A4. $r(t, \tau) = \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_1)|t_1 = \tau\right)$ is uniformly continuous in the interior of $\mathcal{T}$ and bounded in $\mathcal{T}$.

A5. $\mathrm{v}_0(\tau) = \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(\tau)\right) w_1(\mathbf{x}_1)|t_1 = \tau\right)$ and $\mathrm{v}_1(\tau) = \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(\tau)\right) \mathbf{x}_1 w_1(\mathbf{x}_1)|t_1 = \tau\right)$ are uniformly continuous in the interior of $\mathcal{T}$ and $\mathcal{I}_{\mathrm{v}_0} = \inf_{t\in[0,1]} |\mathrm{v}_0(\tau)| > 0$.

A6. Ψ, χ, χ_1, w_1 and $\psi_1(\mathbf{x}) = \mathbf{x}w_1(\mathbf{x})$ are bounded functions.

A7. K is a function of bounded variation with compact support $[0, 1]$ and it satisfies $\int K(u)du = 1$ and $\int u\, K(u)du = 0$.

A8. The bandwidth sequence satisfies $nh_n^3/\log(n) \to \infty$ and $n^{\frac{1}{2}}h_n^4\log(n) \to 0$.

Theorem 4.1. *Assume that **A1** to **A8** hold. Moreover, assume that*

a) $\mathcal{G} = \left\{g(y, \mathbf{x}, u) = \chi\left(y, \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}) - \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_1)|t_1 = u\right), a \in \mathbb{R}\right\}$, *has covering number* $N(\epsilon, \mathcal{G}, L^1(\mathbb{Q})) \leq A\epsilon^{-W}$, *for any probability* $\mathbb{Q}$ *and* $0 < \epsilon < 1$.

b) $\psi_{1,2}(\mathbf{x}) = w_1(\mathbf{x})\|\mathbf{x}\|^2$ *is bounded or* $\sup_{t\in\mathcal{T}} \mathbb{E}_0\left(\psi_{1,2}(\mathbf{x})|t\right) < \infty$.

Then, under $H_0 : \eta \in \{\alpha + \gamma\, t,\ \alpha \in \mathbb{R}, \gamma \in \mathbb{R}\}$, *we have that* $v_n^{-1}\left(\mathcal{T}_1 - m_n\right) \xrightarrow{w} N(0, 1)$, *with* $m_n = c_{1,\Psi}\, h_n^{-1}\int K^2(u)du$ *and* $v_n^2 = 2c_{2,\Psi}\, h_n^{-1}\int \left(K * K(u)\right)^2 du$, *where*

$$\begin{aligned}
c_{1,\Psi} &= \mathbb{E}\left(w(t_1)\mathbb{E}\left[w_2(\mathbf{x}_i)\frac{H'\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_1\right)^2}{V\left(H\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_1\right)\right)}\Big|t_1\right] \mathbb{E}\left[w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1, t_1)\Big|t_1\right] \mathrm{v}_0(t_1)^{-2} f(t_1)^{-1}\right) \\
c_{2,\Psi} &= \mathbb{E}\left(\left[\mathbb{E}\left\{w_2(\mathbf{x}_i)\frac{H'\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_1\right)^2}{V\left(H\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_1\right)\right)}\Big|t_1\right\}\right]^2 \left[\mathbb{E}\left\{w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1, t_1)\Big|t_1\right\}\right]^2 \frac{w^2(t_1)}{\mathrm{v}_0(t_1)^4 f(t_1)}\right) \\
\sigma^2(\mathbf{x}_0, t_0) &= \mathbb{E}\left\{\left[\Psi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)\right) - \mathbb{E}_0\left(\Psi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)\right)|(\mathbf{x}_1, t_1)\right)\right]^2 |(\mathbf{x}_1, t_1) = (\mathbf{x}_0, t_0)\right\}.
\end{aligned}$$

Remark 4.1. When considering the canonical exponential family described in the Introduction $H'\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)\right)=V\left(H\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)\right)\right)$ and so

$$\begin{aligned} c_{1,\Psi} &= \mathbb{E}\left(w(t_1)\mathbb{E}\left[w_2(\mathbf{x}_1)H'\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t_1\right)\Big|t_1\right]\mathbb{E}\left[w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1,t_1)\Big|t_1\right]\mathrm{v}_0(t_1)^{-2}f(t_1)^{-1}\right) \\ c_{2,\Psi} &= \mathbb{E}\left(w^2(t_1)\left[\mathbb{E}\left\{w_2(\mathbf{x}_1)H'\left(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t_1\right)\Big|t_1\right\}\right]^2\left[\mathbb{E}\left\{w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1,t_1)\Big|t_1\right\}\right]^2\frac{1}{\mathrm{v}_0(t_1)^4 f(t_1)}\right) . \end{aligned}$$

5 A Montecarlo test

In this section, we develop a boostrap procedure to implement the goodness–of–fit test for linearity. The need of bootstraping has been studied by several authors such as Mammen and Härdle (1993), Härdle *et al.* (1998). These authors applied a wild bootstrap procedure to construct the bootstrap samples. However, in the present setting due to the expensive computing time needed to compute the robust estimators, a linearized Montecarlo as defined in Zhu (2005) provides a better approach. This approach was also considered in Zhu and Zhang (2004) who propose a resampling procedure for approximating the $p-$value when considering a log-likelihood ratio test statistics for testing homogeneity. Rémillard and Scaillet (2009) and Kojadinovic and Yan (2011) applied this method to provide fast goodness–of–fit tests for copulas.

As it will be shown in the Appendix, $\mathcal{T}_1=R_n+O_p((n/h)^{\frac{1}{2}}\rho_n\log n)$, under $H_0:\eta\in\{\alpha+\gamma\, t,\ \alpha\in\mathbb{R},\gamma\in\mathbb{R}\}$, with

$$\begin{aligned} R_n &= \sum_{i=1}^n w(t_i)w_2(\mathbf{x}_i)\frac{H'\left(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t_i\right)^2}{V\left(H\left(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t_i\right)\right)}\mathrm{v}_0(t_i)^{-2}f(t_i)^{-2}\times \\ &\quad \left\{\sum_{j=1}^n W_{0,j}(t_i)w_1(\mathbf{x}_j)\left[\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_i)\right)-\mathbb{E}_0\left(\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_i)\right)|(\mathbf{x}_j,t_j,\mathbf{x}_i,t_i)\right)\right]\right\}^2 \\ &= n\int w(t)w_2(\mathbf{x})\frac{H'\left(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t\right)^2}{V\left(H\left(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t\right)\right)}\mathrm{v}_0(t)^{-2}f(t)^{-2}\mathcal{W}_n^2(t)dF_n(\mathbf{x},t) , \end{aligned}$$

where $\mathcal{W}_n(t)=\sum_{j=1}^n W_{0,j}(t)w_1(\mathbf{x}_j)\left[\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)-\mathbb{E}_0\left(\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_i)\right)|(\mathbf{x}_j,t_j,t_i=t)\right)\right]$. This suggests the following Montecarlo procedure

Step B1 Given a sample $\{(y_i,\mathbf{x}_i,t_i)\}_{1\le i\le n}$ compute the estimators $(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0},\widehat{\alpha}_{\mathrm{H}_0},\widehat{\gamma}_{\mathrm{H}_0})$ as in (3).

Define

- $\widehat{\mathrm{v}}_0(t)=\sum_{i=1}^n W_i(t)\chi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{H_0}(t)\right)w_1(\mathbf{x}_i)$ with $\widehat{\eta}_{H_0}(t)=\widehat{\alpha}_{\mathrm{H}_0}+\widehat{\gamma}_{\mathrm{H}_0}t$.
- $\widehat{\epsilon}_j(t)=\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{H_0}(t)\right)-\mathbb{E}_0\left(\Psi\left(y_j,\mathbf{x}_j^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{H_0}(t)\right)|(\mathbf{x}_j,t_j)\right)$

Step B2 Generate n random variables $\epsilon_1^\star\ldots\epsilon_n^\star$, independent of the sample $\{(y_i,\mathbf{x}_i,t_i)\}_{1\le i\le n}$ and such that $\mathbb{E}(\epsilon_i^\star)=0$, $\mathrm{VAR}(\epsilon_i^\star)=1$ and $\epsilon_i^\star$ are bounded. For instance, we generate n observations from the two point distribution $P^\star(\epsilon^\star=a)=p$ and $P^\star(\epsilon^\star=b)=1-p$, with $a=(1-\sqrt{5})/2$, $b=(1+\sqrt{5})/2$ and $p=(5+\sqrt{5})/10$.

Step B3 Define $R_n^\star=R_n^\star(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0},\widehat{\mathrm{v}}_0,\widehat{\eta}_{H_0})$ with

$$R_n^\star = \sum_{i=1}^n w(t_i)w_2(\mathbf{x}_i)\frac{H'\left(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{H_0}(t_i)\right)^2}{V\left(H\left(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{H_0}(t_i)\right)\right)}\widehat{\mathrm{v}}_0(t_i)^{-2}\left\{\sum_{j=1}^n W_j(t_i)w_1(\mathbf{x}_j)\widehat{\epsilon}_j(t_i)\epsilon_j^\star\right\}^2 .$$

Step B4 Repeat **Step B2** and **Step B3** N_{boot} times, to get N_{boot} values of $R_n^\star$, say $R_{n,i}^\star$, $1\le i\le N_{boot}$.

The $(1-\alpha)-$quantiles of the distribution of R (an so of $\mathcal{T}_1$) can be approximated by the $(1-\alpha)-$quantiles of the conditional distribution of $R^\star$. The $p-$value can be estimated by $\widehat{p} = k/(N_{boot}+1)$ where k is the number of $R^\star_{n,i}$ which are larger or equal than $\mathcal{T}_1$.

6 Monte Carlo study

This section contains the results of a preliminary simulation study conducted with the aim of comparing the performance of the proposed testing procedure with the classical one. We have considered a logistic partly linear model. The robust estimators correspond to those controling large values of the deviance and they were computed using the score function defined in Croux and Haesbroeck (2002) with tuning constant $c = 0.5$. The weight functions w_1 and w_2 used to control high leverage points were taken as the Tukey's biweight function with tuning constant $c = 4.685$, i.e,

$$w_1^2(x_i) = w_2^2(x_i) = \begin{cases} \left(1 - \left[\dfrac{x_i - M_n}{4.685}\right]^2\right)^2 & |x_i - M_n| \leq 4.685 \\ 0 & |x_i - M_n| \geq 4.685 \end{cases}$$

with M_n the median of x_i since we have considered $x_i \in \mathbb{R}$. The central model denoted C_0 in tables and figures corresponds to select $x_i \sim \mathcal{U}(-1,1)$ and $t_i \sim \mathcal{U}(0,1)$, independent each other.

On the other hand, the response variable was generated as

$$y_i = \begin{cases} 1, & \beta_0 x_i + \eta_0(t_i) + \epsilon_i \geq 0 \\ 0, & \beta_0 x_i + \eta_0(t_i) + \epsilon_i < 0 \end{cases}$$

with $\beta_0 = 2$, $\eta_0(t,\Delta) = (t-0.5) + \Delta\cos(6\pi(t-0.5))$, ϵ_i a logistic random variable, i.e, $y|(x,t) \sim Bi(1, p(x,t))$ with $\log\left(p(x,t)/\left(1-p(x,t)\right)\right) = 2x + (t-0.5) + \Delta\cos(6\pi(t-0.5))$, that is, $H(u) = \exp(u)/(1+\exp(u))$. The value $\Delta = 0$ corresponds to the null hypothesis, $H_0 : \eta \in \{\alpha + \gamma\, t,\ \alpha \in \mathbb{R}, \gamma \in \mathbb{R}\}$, while as alternatives we have considered a grid of 10 equally spaced values of $\Delta \in [0.2, 2.0]$. We have considered $NR = 1000$ replications of samples of size $n = 200$ with bandwidth $h = 0.1$ and $N_{boot} = 5000$ bootstrap samples. The Epanechinikov kernel $K(t) = (3/4)(1-t^2)I_{[-1,1]}(t)$ was selected for the smoothing procedure.

Figure 1 gives the frequency of rejection both for the classical and robust procedure for the uncontaminated samples. The nominal level was 0.10. The frequency of rejection of the asymptotic test is plotted in lines combined with filled diamonds while that of the Montecarlo test corresponds to the solid line. As expected the Montecarlo test improves the performance of the asymptotic ones, for the sample size considered.

a) b)

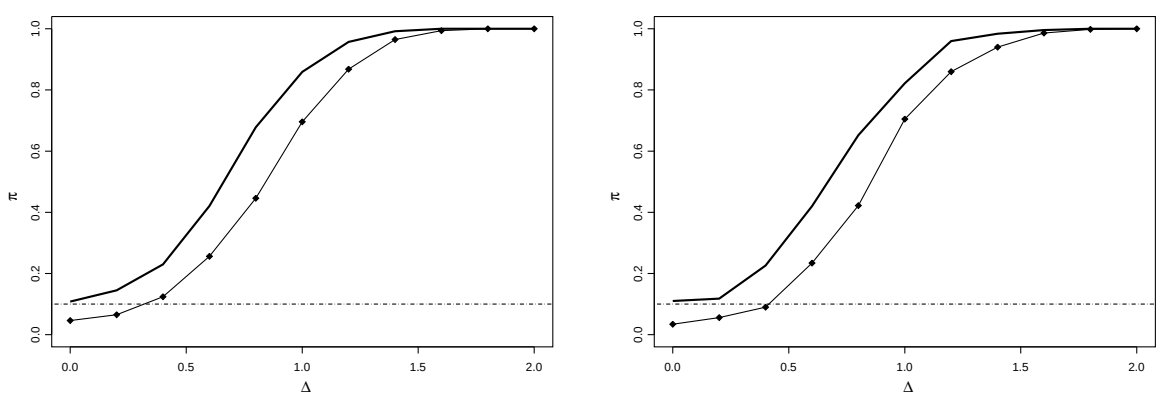

Figure 1: Frequency of rejection π of the asymptotic test, plotted with filled diamonds, and the Montecarlo test plotted with a solid line. a) Classical test b) Robust test.

For each generated sample, we have considered the following contaminations labeled C_1 and C_2 in tables and figures. We have first generated a sample $u_i \sim \mathcal{U}(0,1)$ for $1 \leq i \leq n$ and then, the contaminated sample, denoted $(y_{i,c}, x_{i,c}, t_i)$, is defined as follows for each contamination scheme

- Contamination C_1 introduces *bad* high leverage points in the carriers x, without changing the responses already generated, i.e., $(y_{i,c}, x_{i,c}) = (y_i, x_i)$ if $u_i \leq 0.90$ and $(y_{i,c}, x_{i,c}) = (y_i, x_{i,new})$ if $u_i > 0.90$, where $x_{i,new}$ is a new observation from a $\mathrm{N}(10,1)$.
- Contamination C_2 outlying observations in the responses generated according to the model but with an incorrect carrier x. Let $x_{i,new}$ a new observation from a $N(-10, 1/6)$ and $p_{i,new} = H(2x_{i,new} + \eta_0(t_i, \Delta)$, define $y_{i,new} \sim Bi(1, x_{i,new})$. Then, $(y_{i,c}, x_{i,c}) = (y_i, x_i)$ if $u_i \leq 0.90$ and $(y_{i,c}, x_{i,c}) = (y_{i,new}, x_{i,new})$ if $u_i > 0.90$.

Figures 2 and 3 give the frequency of rejection both for the classical and robust procedure for the contaminated samples. Figure 2 reports the frequencies of rejection for both the asymptotic and Montecarlo procedure, on the other hand, only the results for the Montecarlo test are reported for C_2 since the asymptotic ones behave similarly. The results show that the classical test seem to be quite unsensitive to high leverage points if the model is adequate, while it is sensitive to a missleading model.

P Appendix. Proofs

In this section we will give the proof of Theorem 4.1. We will need the following lemmas

Lemma P.1. *Assume that **A1** to **A4**, **A6** and **A7** hold and that $nh_n^3/\log(n) \to \infty$. Then,*

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\leq\rho_n}} \left|S_n^{(0,1)}(\boldsymbol{\beta}, \eta_0(t), t)\right| = O_p\left(\rho_n\sqrt{\log n}\right)$$

Proof. Using that χ and ψ_1 are bounded functions it is easy to see that

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\leq\rho_n}} \left|S_n^{(0,1)}(\boldsymbol{\beta}, \eta_0(t), t) - S_n^{(0,1)}(\boldsymbol{\beta}_0, \eta_0(t), t)\right| \leq \|\chi\|_\infty \|\psi_1\|_\infty \rho_n \sup_{t\in[2h_n,1-2h_n]} \sum_{i=1}^n |W_{0,i}(t)| = O_p(\rho_n) \ .$$

Note that $\mathbb{E}_0\left(\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_i)\right) w_1(\mathbf{x}_i)|t_i\right) = 0$. Denote

$$\begin{aligned} v_{0,n}^1(t) &= \sum_{i=1}^n W_{0,i}(t)\left[\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_i) - \mathbb{E}_0\left(\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_i)|t_i\right)\right] \\ A_n^1(t) &= \sum_{i=1}^n W_{0,i}(t)\mathbb{E}_0\left(\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_i)|t_i\right) \end{aligned}$$

with $W_{0,i}(t)$ defined in (2).

It will be enough to show that

$$\sup_{t\in[2h_n,1-2h_n]} \left|v_{0,n}^1(t)\right| = O_p(h_n^2) = O_p\left(\rho_n\sqrt{\log n}\right) \tag{P.1}$$

$$\sup_{t\in[2h_n,1-2h_n]} \left|A_n^1(t)\right| = O_p(h_n^2) = O_p\left(\rho_n\sqrt{\log n}\right) \tag{P.2}$$

We begin by proving (P.2). Define $m(\tau) = \mathbb{E}_0\left(\Psi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta} + \eta_0(t)\right) w_1(\mathbf{x}_1)|t_1 = \tau\right)$ and $g(u) = m(u)f(u)$. Then, we have that $m(t) = 0$ and we can write $A_n^1(t) = \sum_{i=1}^n W_{0,i}(t)m(t_i)$. On the other hand, using that K has bounded support, $\int uK(u)du = 0$ and $t \in [2h_n, 1-2h_n]$, we get that

$$\mathbb{E}(A_n^1(t)) = \frac{1}{h_n}\int K\left(\frac{u-t}{h_n}\right) g(u)du = g(t)\int K(u)\,udu + h_n g'(t)\int K(u)\,udu + h_n^2\int K(u)\,u^2 g''(\xi_n)du \ ,$$

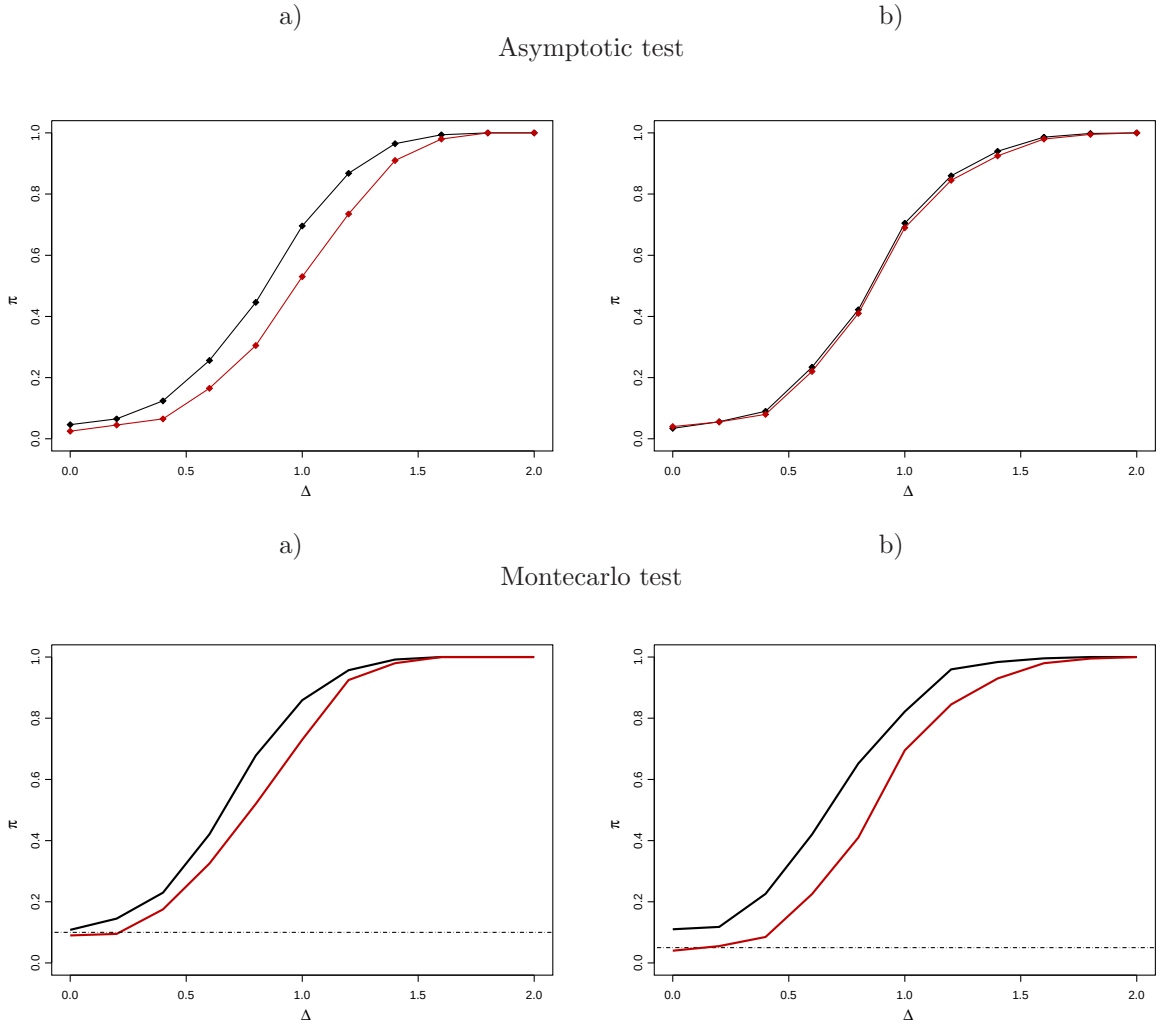

Figure 2: Frequency of rejection π of the asymptotic and Montecarlo test, under C_0 in black and under C_1 in red. a) Classical test b) Robust test

which implies that $\sup_{t\in[2h_n,1-2h_n]} |\mathbb{E}(A_n^1(t))| = O(h_n^2)$. Now, since K is a bounded variation kernel and f and r are bounded, Theorem 37 in Pollard (1984) with $\mathcal{F}_n = \{f_{t,h_n}(u) = K\left((u-t)/h_n\right)\ m(u),\ t \in [2h_n, 1-2h_n]\}$, $\delta_n^2 = h_n$, $\alpha_n = h_n$ and the fact that $nh_n^3/\log(n) \to \infty$, entails that $\sup_{t\in[2h_n,1-2h_n]} |A_n^1(t) - \mathbb{E}(A_n^1(t))| = o_p(h_n^2)$

Let us prove (P.1). Note that $\mathbb{E}_0\left(v_{0,n}^1(t)\right) = 0$. Denote

$$\begin{aligned}\widetilde{\mathcal{F}}_n &= \{f_{t,h_n}(y,\mathbf{x},u) = (2\|K\|_\infty\|\Psi\|_\infty\|w_1\|_\infty)^{-1} K\left(\frac{u-t}{h_n}\right)\left[\Psi\left(y, \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x})\right.\\ &- \left.\mathbb{E}_0\left(\Psi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_1)|t_1 = u\right)\right], \quad t \in [2h_n, 1-2h_n]\}.\end{aligned}$$

Using that χ and η_0 are bounded, we obtain that the family of functions

$$\mathcal{G}_n = \{g_t(y,\mathbf{x},u) = \Psi\left(y, \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}) - \mathbb{E}_0\left(\Psi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) w_1(\mathbf{x}_1)|t_1 = u\right), \quad t \in [2h_n, 1-2h_n]\},$$

has covering number $N(\epsilon, \mathcal{G}_n, L^1(\mathbb{Q})) \leq A\epsilon^{-W}$, for any probability $\mathbb{Q}$ and $0 < \epsilon < 1$ which together with the fact that K has bounded variation and Problem 27 in Pollard (1984) allow to conclude that $N(\epsilon, \widetilde{\mathcal{F}}_n, L^1(\mathbb{Q})) \leq B\epsilon^{-C}$,

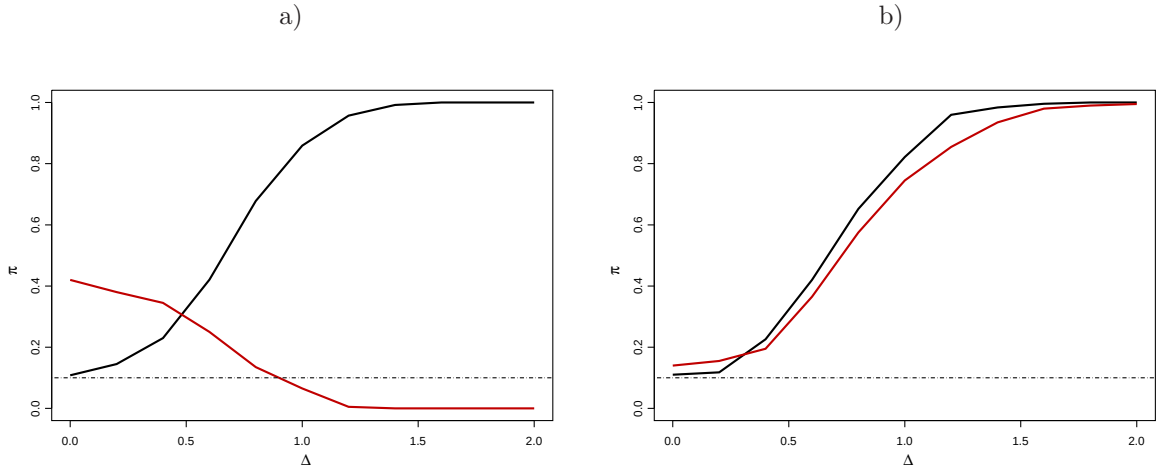

Figure 3: Frequency of rejection π of the Montecarlo test, under C_0 in black and under C_2 in red. a) Classical test b) Robust test

for some constants B and C not depending on n. On the other hand, $|f_{t,h_n}(y,\mathbf{x},u)| \leq 1$ and $\text{VAR}(f_{t,h_n}(y_1,\mathbf{x}_1,t_1)) \leq h_n \|K\|_\infty^{-2} \|f\|_\infty \int K^2(u)du$. Therefore, Theorem 37 in Pollard (1984) with $\delta_n^2 = h_n$, $\alpha_n = h_n$ and the fact that $nh_n^3/\log(n) \to \infty$, concludes the proof of (P.1). $\square$

Lemma P.2. *Assume that* ***A1*** *to* ***A7*** *hold and that* $nh_n^3/\log(n) \to \infty$. *If the family of functions* $\mathcal{G} = \{g(y,\mathbf{x},u) = \chi\left(y, \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}) - \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_1)|t_1 = u\right), a \in \mathbb{R}\}$, *has covering number* $N(\epsilon, \mathcal{G}, L^1(\mathbb{Q})) \leq A\epsilon^{-W}$, *for any probability* $\mathbb{Q}$ *and* $0 < \epsilon < 1$, *we have that*

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\leq\rho_n}} \left|\widehat{\eta}_{\boldsymbol{\beta}}(t) - \eta_0(t)\right| = O_p\left(\rho_n\sqrt{\log n}\right)$$

PROOF. Using that $S_n^{(0,1)}(\boldsymbol{\beta}, \widehat{\eta}_{\boldsymbol{\beta}}(t), t) = 0$, a Taylor's expansion of order one, leads to

$$\widehat{\eta}_{\boldsymbol{\beta}}(t) - \eta_0(t) = -S_n^{(0,1)}(\boldsymbol{\beta}, \eta_0(t), t)\left[\frac{\partial}{\partial a} S_n^{(0,1)}(\boldsymbol{\beta}, a, t)\Big|_{a=\xi(t)}\right]^{-1}$$

with $\xi(t)$ an intermediate point between $\widehat{\eta}_{\boldsymbol{\beta}}(t)$ and $\eta_0(t)$. Note that

$$\frac{\partial}{\partial a} S_n^{(0,1)}(\boldsymbol{\beta}, a, t) = \sum_{i=1}^n W_{0,i}(t)\chi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + a\right) w_1(\mathbf{x}_i)$$

Using that χ_1 and ψ_1 are bounded functions analogous arguments to those given in Lemma P.1, allow to show that

$$\sup_{\substack{t\in[2h_n,1-2h_n],a\in\mathbb{R}\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\leq\rho_n}} \left|\sum_{i=1}^n W_{0,i}(t)\left[\chi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} + a\right) - \chi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right)\right] w_1(\mathbf{x}_i)\right| \quad = \quad O_p(\rho_n)$$

and

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ a\in\mathbb{R}}} \left|\sum_{i=1}^n W_{0,i}(t)\left[\chi\left(y_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_i) - \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_1)|t_1 = t_i\right)\right]\right| = o_p(1)$$

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ a\in\mathbb{R}}} \left|\sum_{i=1}^n W_i(t)\mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_1)|t_1 = t_i\right) - \mathbb{E}_0\left(\chi\left(y_1, \mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + a\right) w_1(\mathbf{x}_1)|t_1 = t\right)\right| = o_p(1)$$

which together with **A2** and **A5** entail that

$$\inf_{\substack{t\in[2h_n,1-2h_n],a\in\mathbb{R}\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\rho_n}}\left|\frac{\partial}{\partial a}S_n^{(0,1)}(\boldsymbol{\beta},a,t)\Big|_{a=\xi(t)}\right| > \inf_{t\in[0,1]} f(t)\,\mathcal{I}_{\mathrm{v}_0}/2\ ,$$

with probability converging to 1. The conclusion follows now from Lemma P.1. □

Lemma P.3. *Assume that **A1** to **A7** hold and that $nh_n^3/\log(n)\to\infty$. If, in addition, $\psi_{1,2}(\mathbf{x})=w_1(\mathbf{x})\|\mathbf{x}\|^2$ is bounded or $\sup_{t\in\mathcal{T}}\mathbb{E}_0\left(\psi_{1,2}(\mathbf{x})|t\right)<\infty$ and the class of functions $\mathcal{G}=\{g(y,\mathbf{x},u)=\chi\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+a\right)w_1(\mathbf{x})-\mathbb{E}_0\left(\chi\left(y_1,\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+a\right)w_1(\mathbf{x}_1)|t_1=u\right),\quad a\in\mathbb{R}\}$, has covering number $N(\epsilon,\mathcal{G},L^1(\mathbb{Q}))\le A\epsilon^{-W}$, for any probability $\mathbb{Q}$ and $0<\epsilon<1$, we have that*

$$\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\rho_n}}\left|\widehat{\eta}_{\boldsymbol{\beta}}(t)-\widehat{\widehat{\eta}}(t)-\widehat{R}(\boldsymbol{\beta},t)\right|=O_p(\rho_n^2\log n) \tag{P.3}$$

with

$$\widehat{\widehat{\eta}}(t) = \eta_0(t)-\{\mathrm{v}_0(t)f(t)\}^{-1}\sum_{i=1}^n W_{0,i}(t)w_1(\mathbf{x}_i)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right) \tag{P.4}$$

$$\widehat{R}(\boldsymbol{\beta},t) = \mathrm{v}_0(t)^{-1}\mathrm{v}_1(t)^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)\ . \tag{P.5}$$

Proof. $\widehat{\eta}_{\boldsymbol{\beta}}(t)$ satisfies $S_n^{(0,1)}(\boldsymbol{\beta},\widehat{\eta}_{\boldsymbol{\beta}}(t),t)=0$ and so, using a Taylor expansion, we get

$$\begin{aligned}
0 &= \sum_{i=1}^n W_{0,i}(t)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}+\widehat{\eta}_{\boldsymbol{\beta}}(t)\right)w_1(\mathbf{x}_i)=\sum_{i=1}^n W_{0,i}(t)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\\
&+ \sum_{i=1}^n W_{0,i}(t)\chi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]\\
&+ \frac{1}{2}\sum_{i=1}^n W_{0,i}(t)\chi_1\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]^2
\end{aligned}$$

with $\boldsymbol{\beta}^*$ and $\eta^*(t)$ intermediate points. Then, we have that

$$\begin{aligned}
\mathrm{v}_{0,n}(t)f_n(t)\left(\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right) &= -\sum_{i=1}^n W_{0,i}(t)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)-f_n(t)\,\mathrm{v}_{1,n}(t)^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)\\
&- \frac{1}{2}\sum_{i=1}^n W_{0,i}(t)\chi_1\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]^2
\end{aligned}$$

with $\mathrm{v}_{0,n}(t)=\sum_{i=1}^n W_i(t)\chi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)$, $\mathrm{v}_{1,n}(t)=\sum_{i=1}^n W_i(t)\chi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\mathbf{x}_i$ and $f_n(t)=\sum_{i=1}^n W_{0,i}(t)$, which implies that

$$\begin{aligned}
\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t) &= -\mathrm{v}_{0,n}(t)^{-1}\left[f_n(t)^{-1}\sum_{i=1}^n W_{0,i}(t)w_1(\mathbf{x}_i)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)+\mathrm{v}_{1,n}(t)^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)\right]\\
&- \frac{1}{2}\,\mathrm{v}_{0,n}(t)^{-1}\sum_{i=1}^n W_i(t)\chi_1\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]^2
\end{aligned}$$

Using that χ_1 is bounded, $\psi_{1,2}(\mathbf{x})=w_1(\mathbf{x})\|\mathbf{x}\|^2$ is bounded or $\sup_{t\in\mathcal{T}}\mathbb{E}_0\left(\psi_{1,2}(\mathbf{x})|t\right)<\infty$, Lemma P.2 entails that

$$\begin{aligned}
&\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\rho_n}}\left|\sum_{i=1}^n W_i(t)\chi_1\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\boldsymbol{\beta}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]^2\right|\\
\le\ & 2\|\chi_1\|_\infty\sup_{\substack{t\in[2h_n,1-2h_n]\\ \|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\rho_n}}\left[\sum_{i=1}^n|W_i(t)|w_1(\mathbf{x}_i)\|\mathbf{x}_i\|^2\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|^2+\|w_1\|_\infty\sum_{i=1}^n|W_i(t)|\left[\widehat{\eta}_{\boldsymbol{\beta}}(t)-\eta_0(t)\right]^2\right]\\
=\ & O_p(\rho_n^2\log n)
\end{aligned}$$

Then, using that **A2** holds and the fact that

$$\inf_{t\in[2h_n,1-2h_n]}|f_n(t)| \geq \inf_{t\in[2h_n,1-2h_n]} f(t) + \sup_{t\in[2h_n,1-2h_n]}|f_n(t)-f(t)|$$

it will be enough to show that

$$\sup_{t\in[2h_n,1-2h_n]}|\mathrm{v}_{0,n}(t)-\mathrm{v}_0(t)| = O_p(\rho_n\sqrt{\log n}) \tag{P.6}$$

$$\sup_{t\in[2h_n,1-2h_n]}|\mathrm{v}_{1,n}(t)-\mathrm{v}_1(t)| = O_p(\rho_n\sqrt{\log n}) \tag{P.7}$$

$$\sup_{t\in[2h_n,1-2h_n]}|\sum_{i=1}^n W_{0,i}(t)w_1(\mathbf{x}_i)\Psi\left(y_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)| = O_p(\rho_n\sqrt{\log n}) \tag{P.8}$$

$$\sup_{t\in[2h_n,1-2h_n]}|f_n(t)-f(t)| = O_p(h_n^2) = O_p(\rho_n\sqrt{\log n}) \tag{P.9}$$

where $\mathrm{v}_0(t)$ is defined in **A5**. Equation (P.8) follows from Lemma P.1. On the other hand, (P.6), (P.7) and (P.9) follow using analogous arguments to those considered to prove (P.1) to (P.2) together with **A5** and the fact that $\int u\,K(u)du=0$ and $\int K(u)du=1$. Therefore, we have that (P.3) holds concluding the proof. □

The following lemma follows using similar arguments as those considered in Lemma P.3, however, we give its proof for the sake of completeness.

Lemma P.4. *Assume that H_0 holds, i.e., $\eta_0(t)=\alpha_0+\gamma_0 t$. Denote $\widetilde{y}_{i,0}=H\left(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\alpha_0+\gamma_0 t_i\right)$ and $\nu(\tau)=\mathbb{E}\left(w_1(\mathbf{x}_1)\widetilde{\zeta}\left(\widetilde{y}_{1,0},\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(\tau)\right)H'\left(H^{-1}\left(\widetilde{y}_{1,0}\right)\right)|t_1=\tau\right)$ where $\widetilde{\zeta}(y,a)=\partial\widetilde{\Psi}\left(y,a\right)/\partial y$. Under **A1** to **A7** if in addition $nh_n^3/\log(n)\to\infty$, we have that*

$$\sup_{t\in[2h_n,1-2h_n]}\Big|\widehat{\eta}_{\mathrm{H}_0}(t)-(\widehat{\alpha}_{\mathrm{H}_0}+\widehat{\gamma}_{\mathrm{H}_0}t-\alpha_0-\gamma_0 t)\,\mathrm{v}_0(t)^{-1}\nu(t)-\eta_0(t) + \mathrm{v}_0(t)^{-1}f(t)^{-1}\sum_{i=1}^n W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_{i,0},\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)\Big| = O_p(\rho_n^2\log n)\,.$$

Proof. $\widehat{\eta}_{\mathrm{H}_0}(t)$ is the solution of $\sum_{i=1}^n W_i(t)\widetilde{\Psi}(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+a)w_1(\mathbf{x}_i)=0$ or equivalently, it satisfies $\sum_{i=1}^n W_{0,i}(t)\widetilde{\Psi}(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{\mathrm{H}_0}(t))w_1(\mathbf{x}_i)=0$ and so, using a Taylor expansion, we get

$$\begin{aligned}
0 &= \sum_{i=1}^n W_{0,i}(t)\widetilde{\Psi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}+\widehat{\eta}_{\mathrm{H}_0}(t)\right)w_1(\mathbf{x}_i) = \sum_{i=1}^n W_{0,i}(t)\widetilde{\Psi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\\
&+ \sum_{i=1}^n W_{0,i}(t)\widetilde{\chi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\mathrm{H}_0}(t)-\eta_0(t)\right]\\
&+ \frac{1}{2}\sum_{i=1}^n W_{0,i}(t)\widetilde{\chi}_1\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\mathrm{H}_0}(t)-\eta_0(t)\right]^2
\end{aligned}$$

with $\widetilde{\chi}(\mu,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a)=\mathbb{E}\left(\chi\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a\right)|(\mathbf{x},t)\right)$, $\widetilde{\chi}_1(\mu,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a)=\mathbb{E}\left(\chi_1\left(y,\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}+a\right)|(\mathbf{x},t)\right)$, where the conditional expectation are taken when $y|(\mathbf{x},t)\sim F(\cdot,\mu)$, and $\boldsymbol{\beta}^*$ and $\eta^*(t)$ intermediate points. Then,

$$\begin{aligned}
\widetilde{\mathrm{v}}_{0,n}(t)f_n(t)\left(\widehat{\eta}_{\mathrm{H}_0}(t)-\eta_0(t)\right) &= -\sum_{i=1}^n W_{0,i}(t)\widetilde{\Psi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)-f_n(t)\,\widetilde{\mathrm{v}}_{1,n}(t)^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}-\boldsymbol{\beta}_0\right)\\
&- \frac{1}{2}\sum_{i=1}^n W_{0,i}(t)\widetilde{\chi}_1\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^*+\eta^*(t)\right)w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}-\boldsymbol{\beta}_0\right)+\widehat{\eta}_{\mathrm{H}_0}(t)-\eta_0(t)\right]^2
\end{aligned}$$

with $\widetilde{\mathrm{v}}_{0,n}(t)=\sum_{i=1}^n W_i(t)\widetilde{\chi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)$, $\widetilde{\mathrm{v}}_{1,n}(t)=\sum_{i=1}^n W_i(t)\widetilde{\chi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)w_1(\mathbf{x}_i)\mathbf{x}_i$ and $f_n(t)=\sum_{i=1}^n W_{0,i}(t)$, which implies that

$$\widehat{\eta}_{\mathrm{H}_0}(t)-\eta_0(t) = -\widetilde{\mathrm{v}}_{0,n}(t)^{-1}\left[f_n(t)^{-1}\sum_{i=1}^n W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_i,\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t)\right)+\widetilde{\mathrm{v}}_{1,n}(t)^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0}-\boldsymbol{\beta}_0\right)\right]$$

$$- \quad \frac{1}{2}\,\widetilde{\mathrm{v}}_{0,n}(t)^{-1}\sum_{i=1}^{n} W_i(t)\widetilde{\chi}_1\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^* + \eta^*(t)\right) w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\right) + \widehat{\eta}_{\mathrm{H}_0}(t) - \eta_0(t)\right]^2$$

Using that χ_1 is bounded, $\psi_{1,2}(\mathbf{x}) = w_1(\mathbf{x})\|\mathbf{x}\|^2$ is bounded or $\sup_{t\in\mathcal{T}}\mathbb{E}_0\left(\psi_{1,2}(\mathbf{x})|t\right) < \infty$, and that $\|\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\| \leq \rho_n$, similar arguments to those considered in Lemma P.2 allow to show that $\sup_{t\in[2h_n,1-2h_n]}|\widehat{\eta}_{\mathrm{H}_0}(t) - \eta_0(t)| = O_p\left(\rho_n\sqrt{\log n}\right)$ which entails that

$$\begin{aligned}
&\sup_{t\in[2h_n,1-2h_n]}\left|\sum_{i=1}^{n} W_i(t)\widetilde{\chi}_1\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^* + \eta^*(t)\right) w_1(\mathbf{x}_i)\left[\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\right) + \widehat{\eta}_{\mathrm{H}_0}(t) - \eta_0(t)\right]^2\right| \\
\leq\ & 2\|\chi_1\|_\infty \sup_{t\in[2h_n,1-2h_n]}\left[\sum_{i=1}^{n}|W_i(t)|w_1(\mathbf{x}_i)\|\mathbf{x}_i\|^2\|\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\|^2 + \|w_1\|_\infty\sum_{i=1}^{n}|W_i(t)|\left[\widehat{\eta}_{\mathrm{H}_0}(t) - \eta_0(t)\right]^2\right] \\
=\ & O_p(\rho_n^2\log n)
\end{aligned}$$

Then, using **A2**, the fact that $\inf_{t\in[2h_n,1-2h_n]}|f_n(t)| \geq \inf_{t\in[2h_n,1-2h_n]} f(t) + \sup_{t\in[2h_n,1-2h_n]}|f_n(t) - f(t)|$ and (P.9) it will be enough to show that

$$\sup_{t\in[2h_n,1-2h_n]}|\widetilde{\mathrm{v}}_{0,n}(t) - \mathrm{v}_0(t)| = O_p(\rho_n\sqrt{\log n}) \tag{P.10}$$

$$\sup_{t\in[2h_n,1-2h_n]}|\widetilde{\mathrm{v}}_{1,n}(t) - \mathrm{v}_1(t)| = O_p(\rho_n\sqrt{\log n}) \tag{P.11}$$

$$\sup_{t\in[2h_n,1-2h_n]}\left|\sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right)\right| = O_p(\rho_n\sqrt{\log n}) \tag{P.12}$$

where $\mathrm{v}_0(t)$ is defined in **A5**. Equation (P.12) follows from Lemma P.1. On the other hand, (P.10), (P.11) and (P.9) follow using similar arguments to those considered to prove (P.1) to (P.2) together with the fact that $\int u, K(u)du = 0$ and $\int K(u)du = 1$. Therefore, we have that

$$\sup_{t\in[2h_n,1-2h_n]}|\widehat{\eta}_{\mathrm{H}_0}(t) - \overline{\eta}(t)| = O_p(\rho_n^2\log n)$$

where

$$\overline{\eta}(t) = \eta_0(t) - \mathrm{v}_0(t)^{-1}f(t)^{-1}\sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right)$$

Using that H is twice continuously differentiable and $\widetilde{\Psi}(y,a)$ is twice continuously differentiable with respect to y with derivatives $\widetilde{\zeta}(y,a) = \partial\widetilde{\Psi}(y,a)/\partial y$ and $\widetilde{\zeta}_1(y,a) = \partial^2\widetilde{\Psi}(y,a)/\partial y^2$ we get that

$$\begin{aligned}
\overline{\eta}(t) &= \eta_0(t) - \mathrm{v}_0(t)^{-1}f(t)^{-1}\sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) \\
&= \eta_0(t) - \mathrm{v}_0(t)^{-1}f(t)^{-1}\left[\sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\Psi}\left(\widetilde{y}_{i,0}, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right)\right. \\
&- \sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\widetilde{\zeta}\left(\widetilde{y}_{i,0}, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) H'\left(H^{-1}\left(\widetilde{y}_{i,0}\right)\right)\left(\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\right) + \widehat{\alpha}_{\mathrm{H}_0} - \alpha_0 + \left(\widehat{\gamma}_{\mathrm{H}_0} - \gamma_0\right)t_i\right) \\
&- \frac{1}{2}\sum_{i=1}^{n} W_{0,i}(t)w_1(\mathbf{x}_i)\left\{\widetilde{\zeta}\left(\widetilde{y}_i^*, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right) H''\left(H^{-1}\left(\widetilde{y}_i^*\right)\right)\right. \\
&\left.\left. + \ \widetilde{\zeta}_1\left(\widetilde{y}_i^*, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)\right)\left[H''\left(H^{-1}\left(\widetilde{y}_i^*\right)\right)\right]^2\right\}\left(\mathbf{x}_i^{\mathrm{T}}\left(\widehat{\boldsymbol{\beta}}_{\mathrm{H}_0} - \boldsymbol{\beta}_0\right) + \widehat{\alpha}_{\mathrm{H}_0} - \alpha_0 + \left(\widehat{\gamma}_{\mathrm{H}_0} - \gamma_0\right)t_i\right)^2\right]
\end{aligned}$$

with $\widetilde{y}_i^* = H\left(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}^* + \alpha^* + \gamma^* t_i\right)$ concluding the proof. □

Proof of Theorem 4.1. In order to derive an expansion for the test statistic note that, uniformly on $t \in [2h_n, 1-2h_n]$ we have

$$\begin{aligned}\widehat{\eta}(t) - \widehat{\eta}_{\mathrm{H}_0}(t) &= \widehat{\widehat{\eta}}(t) + \widehat{R}(\widehat{\boldsymbol{\beta}}, t) - (\widehat{\alpha}_{\mathrm{H}_0} + \widehat{\gamma}_{\mathrm{H}_0} t - \alpha_0 - \gamma_0 t)\, \mathrm{v}_0(t)^{-1} \nu(t) - \eta_0(t) \\ &+ \mathrm{v}_0(t)^{-1} f(t)^{-1} \sum_{i=1}^n W_{0,i}(t) w_1(\mathbf{x}_i) \widetilde{\Psi}\left(\widetilde{y}_{i,0}, \mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t)\right) + O_p(\rho_n^2 \log n)\end{aligned}$$

with $\widehat{\widehat{\eta}}(t)$ and $\widehat{R}(\boldsymbol{\beta}, t)$ defined in (P.4) and (P.5), respectively. Hence,

$$\begin{aligned}\widehat{\eta}(t) - \widehat{\eta}_{\mathrm{H}_0}(t) &= -\mathrm{v}_0(t)^{-1} f(t)^{-1} \sum_{i=1}^n W_{0,i}(t) w_1(\mathbf{x}_i) \left[\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t)\right) - \widetilde{\Psi}\left(\widetilde{y}_{i,0}, \mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t)\right)\right] \\ &+ \widehat{R}(\widehat{\boldsymbol{\beta}}, t) - (\widehat{\alpha}_{\mathrm{H}_0} + \widehat{\gamma}_{\mathrm{H}_0} t - \alpha_0 - \gamma_0 t)\, \mathrm{v}_0(t)^{-1} \nu(t) + O_p(\rho_n^2 \log n) \\ &= -\mathrm{v}_0(t)^{-1} f(t)^{-1} \sum_{i=1}^n W_{0,i}(t) w_1(\mathbf{x}_i) \left[\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t)\right) - \mathbb{E}_0\left(\Psi\left(y_i, \mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t)\right) | (\mathbf{x}_i, t_i)\right)\right] \\ &+ O_p(\sqrt{n}) + O_p(\rho_n^2 \log n)\end{aligned}$$

Therefore, we have the following expression for the test statistic $\mathcal{T}_1 = R + O_p((n/h)^{\frac{1}{2}} \rho_n \log n)$ with

$$\begin{aligned}R &= \sum_{i=1}^n \frac{H'\left(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_i\right)^2}{V\left(H\left(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_i\right)\right)} \left[\widehat{\widehat{\eta}}(t_i) - \mathbb{E}\left(\widehat{\widehat{\eta}}(t_i) | \mathbf{x}_1, t_1, \ldots, \mathbf{x}_n, t_n\right)\right]^2 w(t_i) \\ &= \sum_{i=1}^n w(t_i) w_2(\mathbf{x}_i) \frac{H'\left(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_i\right)^2}{V\left(H\left(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \alpha_0 + \gamma_0 t_i\right)\right)} \mathrm{v}_0(t_i)^{-2} f(t_i)^{-2} \mathcal{W}_n^2(t_i) \\ \mathcal{W}_n(t_i) &= \sum_{j=1}^n W_{0,j}(t_i) w_1(\mathbf{x}_j) \left[\Psi\left(y_j, \mathbf{x}_j^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)\right) - \mathbb{E}_0\left(\Psi\left(y_j, \mathbf{x}_j^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)\right) | (\mathbf{x}_j, t_j, \mathbf{x}_i, t_i)\right)\right]\end{aligned}$$

which is a $U-$statistic. Therefore, using standard arguments as in Härdle and Mammen (1993) it follows that $v_n^{-1}(\mathcal{T}_1 - m_n) \xrightarrow{w} N(0,1)$, with $v_n^2 = 2c_{2,\Psi}\, h_n^{-1} \int (K * K(u))^2\, du$ and $m_n = c_{1,\Psi}\, h_n^{-1} \int K^2(u) du$ where $c_1(\Psi)$, $c_2(\Psi)$ and $\sigma^2(\mathbf{x}_0, t_0)$ are given in Theorem 4.1.

Let us verify the expressions for m_n and v_n. Denote $V_{j,i} = w_1(\mathbf{x}_j)[\Psi(y_j, \mathbf{x}_j^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)) - \mathbb{E}_0(\Psi(y_j, \mathbf{x}_j^{\mathrm{T}} + \boldsymbol{\beta}_0 + \eta_0(t_i)) | (\mathbf{x}_j, t_j, \mathbf{x}_i, t_i))]$, then

$$\begin{aligned}R &= \frac{1}{n^2} \sum_{i=1}^n w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} \sum_{j=1}^n \sum_{\ell=1}^n K_{h_n}(t_j - t_i) K_{h_n}(t_\ell - t_i) V_{j,i} V_{\ell,i} \\ R &= \frac{K^2(0)}{n^2 h_n^2} \sum_{i=1}^n w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} V_{i,i}^2 \\ &+ \frac{2K(0)}{n^2 h_n} \sum_{i=1}^n w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} V_{i,i} \sum_{\ell \neq i} K_{h_n}(t_\ell - t_i) V_{\ell,i} \\ &+ \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} \sum_{\ell \neq i} w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} K_{h_n}(t_j - t_i) K_{h_n}(t_\ell - t_i) V_{j,i} V_{\ell,i} \\ &+ \frac{1}{n^2} \sum_{i=1}^n w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} \sum_{j \neq i} K_{h_n}^2(t_j - t_i) V_{j,i}^2 \\ &= R_1 + R_2 + R_3 + R_4\end{aligned}$$

Using that $nh_n^2 \to \infty$ and that

$$\frac{1}{n} \sum_{i=1}^n w(t_i) w_2(\mathbf{x}_i) v_0^{-2}(t_i) f^{-2}(t_i) \frac{H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))} V_{i,i}^2 \xrightarrow{p} \mathbb{E}_0\left(\frac{w(t_1) w_2(\mathbf{x}_i)}{v_0^2(t_1) f^2(t_1)} \frac{H'(\mathbf{x}_1^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_1))^2}{V(H(\mathbf{x}_1^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_1)))} V_{1,1}^2\right)$$

we get that $R_1 \xrightarrow{p} 0$ and so, $h_n^{1/2} R_1 \xrightarrow{p} 0$.

On the other hand, using that $\mathbb{E}(V_{\ell,i}|(\mathbf{x}_\ell, t_\ell, \mathbf{x}_i, t_i)) = 0$ and $\mathbb{E}(V_{i,i}|(\mathbf{x}_i, t_i)) = 0$ and that $V_{\ell,i}$ and $V_{i,i}$ are conditionally independent, for $\ell \neq i$, we get that $\mathbb{E}(R_2) = 0$.

On the other hand, let $Z_i = w(t_i) v_0^{-2}(t_i) f^{-2}(t_i) w_2(\mathbf{x}_i) V_{i,i} (H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2) / V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i)))$, then, we have that $R_2 = (2K(0))/(n^2 h) \sum_{i \neq \ell} Z_i K_{h_n}(t_\ell - t_i) V_{\ell,i}$, and so,

$$\begin{aligned}
\mathrm{Var}(R_2) &= \frac{2K^2(0)}{n^4 h_n^2} \left\{ \sum_{i=1}^n \sum_{\ell \neq i} \sum_{s \neq i} \mathrm{Cov}(Z_i K_{h_n}(t_\ell - t_i) V_{\ell,i}, Z_i K_{h_n}(t_s - t_i) V_{s,i}) \right. \\
&+ \left. \sum_{i=1}^n \sum_{\ell \neq i} \sum_{j \neq i} \mathrm{Cov}(Z_i K_{h_n}(t_\ell - t_i) V_{\ell,i}, Z_j K_{h_n}(t_i - t_j) V_{i,j}) \right\} \\
&= \frac{2K^2(0)}{n^4 h_n^2} \sum_{\ell \neq 1} \mathrm{Cov}(Z_1 K_{h_n}(t_\ell - t_1) V_{\ell,1}, Z_1 K_{h_n}(t_\ell - t_1) V_{\ell,1}) \\
&+ \frac{2K^2(0)}{n^4 h_n^2} \sum_{\ell \neq 1} \mathrm{Cov}(Z_1 K_{h_n}(t_\ell - t_1) V_{\ell,1}, Z_\ell K_{h_n}(t_1 - t_\ell) V_{1,\ell}) \\
&= \frac{2K^2(0) n(n-1)}{n^4 h_n^2} (C_{1,h} + C_{2,h})
\end{aligned}$$

with $C_{1,h} = \mathbb{E}(Z_1^2 K_{h_n}^2(t_2 - t_1) V_{2,1}^2)$ and $C_{2,h} = \mathrm{Cov}(Z_1 K_{h_n}(t_2 - t_1) V_{2,1}, Z_2 K_{h_n}(t_1 - t_2) V_{1,2})$. Note that,

$$\begin{aligned}
C_{1,h} &= \frac{1}{h_n^2} \mathbb{E}\left(Z_1^2 K^2\left(\frac{t_2 - t_1}{h_n}\right) V_{2,1}^2 \right) = \frac{1}{h_n^2} \mathbb{E}\left(\mathbb{E}(Z_1^2 V_{2,1}^2 | (t_1, t_2)) K^2\left(\frac{t_2 - t_1}{h_n}\right) \right) \\
&= \frac{1}{h_n^2} \mathbb{E}\left(R(t_1, t_2) K^2\left(\frac{t_2 - t_1}{h_n}\right) \right) = \frac{1}{h_n} \int R(t_1, t_1 + u h_n) K^2(u) f(t_1) f(t_1 + uh) du\, dt_1
\end{aligned}$$

Hence, $C_{1,h} = O(1)/h_n$. In a similar way, we get that $C_{2,h} = O(1)/h_n$, which implies that $h_n \mathrm{Var}(R_2) \to 0$ as $n \to \infty$, therefore, $h_n^{1/2} R_2 \xrightarrow{p} 0$.

Write $R_4 = (1/n^2) \sum_{i=1}^n \sum_{j \neq i} W_i K_h^2(t_j - t_i) V_{j,i}$ with

$$W_i = w(t_i) w_2(\mathbf{x}_i)\, H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2 \left\{ v_0^2(t_i) f^2(t_i)\, V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))) \right\}^{-1}$$

then, $\mathbb{E}(R_4) = (1/n) \sum_{j \neq 1} \mathbb{E}(W_1 K_{h_n}(t_j - t_1) V_{j,1}^2) = ((n-1)/n) \mathbb{E}(W_1 K_{h_n}^2(t_2 - t_1) \mathbb{E}(V_{2,1}^2 | (\mathbf{x}_1, t_1, \mathbf{x}_2, t_2)))$. Using the fact that $\mathbb{E}(V_{2,1}|(\mathbf{x}_1, t_1, \mathbf{x}_2, t_2)) = 0$, we get that

$$\mathbb{E}(V_{2,1}^2 | (\mathbf{x}_1, t_1, \mathbf{x}_2, t_2) = \mathrm{Var}(V_{2,1} | (\mathbf{x}_1, t_1, \mathbf{x}_2, t_2) = w_1^2(\mathbf{x}_2) \sigma^2(\mathbf{x}_2, t_2, \mathbf{x}_1, t_1) = w_1^2(\mathbf{x}_2) \sigma^2(\mathbf{x}_2, t_2, t_1) \ .$$

Let $R_{4,1} = ((n-1)/n) \mathbb{E}(W_1 K_{h_n}^2(t_2 - t_1) w_1^2(\mathbf{x}_2) \sigma^2(\mathbf{x}_2, t_2))$ and $R_{4,2} = ((n-1)/n) \mathbb{E}(W_1 K_{h_n}^2(t_2 - t_1) w_1^2(\mathbf{x}_2) [\sigma^2(\mathbf{x}_2, t_2, t_1) - \sigma^2(\mathbf{x}_2, t_2)])$, then, $\mathbb{E}(R_4) = R_{4,1} + R_{4,2}$. Using that $\sigma^2(\mathbf{x}_2, t_2, t_1)$ is Lipschitz, we obtain that

$$|\sigma^2(\mathbf{x}_2, t_2, t_1) - \sigma^2(\mathbf{x}_2, t_2)| < |t_1 - t_2| < h_n \ . \tag{P.13}$$

Now, using that K has compact support in $[-1, 1]$, we get that

$$|R_{4,2}| \leq \frac{n-1}{n h_n} \mathbb{E}\left(|W_1| K^2\left(\frac{t_2 - t_1}{h_n}\right) \right) = \frac{n-1}{n} O(1) \ ,$$

and so, $h_n^{1/2} A_2 \to 0$.

Let $a(t_1) = w(t_1) v_0^{-2}(t_1) f^{-2}(t_1)$ and $b(t_1) = \mathbb{E}\left(w_2(\mathbf{x}_i) H'(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))^2 \left\{ V(H(\mathbf{x}_i^{\mathrm{T}} \boldsymbol{\beta}_0 + \eta_0(t_i))) \right\}^{-1} \middle| t_1 \right)$, then

$$h_n^{1/2} R_{4,1} = \frac{n-1}{n h_n^2} h_n^{1/2} \mathbb{E}\left(K^2\left(\frac{t_2 - t_1}{h_n}\right) w_1^2(\mathbf{x}_2) \sigma^2(\mathbf{x}_2, t_2) a(t_1) b(t_1) \right)$$

Denote by $c(t_2) = \mathbb{E}(w_1^2(\mathbf{x}_2)\sigma^2(\mathbf{x}_2, t_2)|t_2)$. Thus,

$$\begin{aligned} h_n^{1/2} R_{4,1} &= \frac{n-1}{nh_n^2} h_n^{1/2} \int K^2\left(\frac{t_2 - t_1}{h_n}\right) a(t_1)b(t_1)c(t_2)f(t_1)f(t_2)dt_1dt_2 \\ &= \frac{n-1}{nh_n} h_n^{1/2} \int K^2(u)du \int a(t_2)b(t_2)c(t_2)f^2(t_2)dt_2 + o(1) \end{aligned}$$

Using analogous arguments to those considered previously when studying the convergence of R_2, one can easily obtain that $h_n \text{VAR}(R_4) \to 0$. Then,

$$h_n^{1/2}\left[R_4 - \frac{1}{h_n}\int K^2(u)du\, \mathbb{E}(a(t_2)b(t_2)c(t_2)f(t_2))\right] \xrightarrow{p} 0\ ,$$

where

$$\mathbb{E}(a(t_2)b(t_2)c(t_2)f(t_2)) = \mathbb{E}\left(\mathbb{E}(w_1^2(\mathbf{x})\sigma^2(\mathbf{x},t)|t)w(t)v_0^{-2}(t)f^{-1}(t)\mathbb{E}\left(w_2(\mathbf{x})\frac{H'(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t)))}\middle| t\right)\right)\ .$$

Finally, we will study the asymptotic behavior of R_3. The expected value of R_3 is equal 0, and so it is enough to study its variance.

$$\begin{aligned} \text{VAR}(R_3) =& \frac{1}{n^4}\sum_{1\le i,s\le n}\sum_{\substack{j\neq i, j\neq \ell, \ell\neq i\\ m\neq s, m\neq r, r\neq i}} \text{COV}\left(a(t_i)w_2(\mathbf{x}_i)\frac{H'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_i))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_i)))}K_{h_n}(t_j - t_i)K_{h_n}(t_\ell - t_i)\ V_{j,i}V_{\ell,i},\right. \\ & \left. a(t_s)w_2(\mathbf{x}_i)\frac{H'(\mathbf{x}_s^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_s))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_s)))}K_{h_n}(t_m - t_s)K_{h_n}(t_r - t_s)V_{m,s}V_{r,s}\right) \\ =& \frac{n}{n^4}\sum_{s=1}^{n}\sum_{\substack{j\neq 1, j\neq \ell, \ell\neq 1\\ m\neq s, m\neq r, r\neq 1}} \mathbb{E}\left(a(t_1)w_2(\mathbf{x}_i)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}K_{h_n}(t_j - t_1)K_{h_n}(t_\ell - t_1)\ V_{j,1}V_{\ell,1}\times\right. \\ & \left. a(t_s)w_2(\mathbf{x}_i)\frac{H'(\mathbf{x}_s^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_s))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_s)))}K_{h_n}(t_m - t_s)K_{h_n}(t_r - t_s)V_{m,s}V_{r,s}\right) \\ =& \frac{1}{n^3}\sum_{j\neq 1, j\neq \ell, \ell\neq 1}\mathbb{E}\left(a^2(t_1)w_2^2(\mathbf{x}_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^4}{V^2(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}K_{h_n}^2(t_j - t_1)K_{h_n}^2(t_\ell - t_1)\times\right. \\ & \left. w_1^2(\mathbf{x}_j)w_1^2(\mathbf{x}_\ell)\sigma^2(\mathbf{x}_j, t_j, t_1)\sigma^2(\mathbf{x}_\ell, t_\ell, t_1)\right) \\ +& \frac{n-1}{n^3}\sum_{\substack{j\neq 1, j\neq \ell, \ell\neq 1\\ m\neq 1, m\neq r, r\neq 2}}\mathbb{E}\left(a(t_1)a(t_2)w_2(\mathbf{x}_1)w_2(\mathbf{x}_2)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}\frac{H'(\mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_2))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_2)))}\times\right. \\ & K_{h_n}(t_j - t_1)K_{h_n}(t_\ell - t_1)K_{h_n}(t_m - t_2)K_{h_n}(t_r - t_2)\times \\ & \left.\mathbb{E}(V_{j,1}V_{\ell,1}V_{m,2}V_{r,2}|(\mathbf{x}_j, t_j, \mathbf{x}_\ell, t_\ell, \mathbf{x}_m, t_m, \mathbf{x}_r, t_r, \mathbf{x}_1, t_1, \mathbf{x}_2, t_2))\right) \\ =& \frac{(n-1)(n-2)}{n^3}\mathbb{E}\left(a^2(t_1)w_2^2(\mathbf{x}_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^4}{V^2(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}K_{h_n}^2(t_2 - t_1)K_{h_n}^2(t_3 - t_1)\times\right. \\ & \left. w_1^2(\mathbf{x}_3)w_1^2(\mathbf{x}_2)\sigma^2(\mathbf{x}_2, t_2, t_1)\sigma^2(\mathbf{x}_3, t_3, t_1)\right) \\ +& 2\frac{(n-1)^2(n-2)}{n^3}\mathbb{E}\left(a(t_1)a(t_2)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}w_2(\mathbf{x}_1)w_2(\mathbf{x}_2)\frac{H'(\mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_2))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_2)))}\times\right. \\ & \left. K_{h_n}(t_3 - t_1)K_{h_n}(t_4 - t_1)K_{h_n}(t_3 - t_2)K_{h_n}(t_4 - t_2)\mathbb{E}(V_{3,1}V_{3,2}|(\mathbf{x}_3, t_3, t_2, t_1))\mathbb{E}(V_{4,1}V_{4,2}|(\mathbf{x}_4, t_4, t_2, t_1))\right) \end{aligned}$$

Hence, $\text{VAR}(R_3) = A_1 + A_2 + A_3$ where

$$A_1 = \frac{(n-1)(n-2)}{n^3}\mathbb{E}\left(a^2(t_1)w_2^2(\mathbf{x}_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1))^4}{V^2(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0 + \eta_0(t_1)))}K_{h_n}^2(t_2 - t_1)K_{h_n}^2(t_3 - t_1)\times\right.$$

$$
\begin{aligned}
& w_1^2(\mathbf{x}_3)w_1^2(\mathbf{x}_2)\sigma^2(\mathbf{x}_2,t_2)\sigma^2(\mathbf{x}_3,t_3)\Big) \\
A_2 &= \frac{(n-1)(n-2)}{n^3}\mathbb{E}\left(a^2(t_1)w_2^2(\mathbf{x}_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1))^4}{V^2(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)))}K_{h_n}^2(t_2-t_1)K_{h_n}^2(t_3-t_1)\right. \\
& \left. w_1^2(\mathbf{x}_3)w_1^2(\mathbf{x}_2)[\sigma^2(\mathbf{x}_2,t_2,t_1)\sigma^2(\mathbf{x}_3,t_3,t_1)-\sigma^2(\mathbf{x}_2,t_2)\sigma^2(\mathbf{x}_3,t_3)]\right) \\
A_3 &= 2\frac{(n-1)^2(n-2)}{n^3}\mathbb{E}\left(a(t_1)a(t_2)w_2(\mathbf{x}_1)w_2(\mathbf{x}_2)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)))}\frac{H'(\mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_2))^2}{V(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_2)))}\times\right. \\
& \left. K_{h_n}(t_3-t_1)K_{h_n}(t_4-t_1)K_{h_n}(t_3-t_2)K_{h_n}(t_4-t_2)\mathbb{E}(V_{3,1}V_{3,2}|(\mathbf{x}_3,t_3,t_2,t_1))\mathbb{E}(V_{4,1}V_{4,2}|(\mathbf{x}_4,t_4,t_2,t_1))\right).
\end{aligned}
$$

Let $b_H(t_1)=\mathbb{E}\left(w_2^2(\mathbf{x}_1)\frac{H'^4(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1))}{V^2(H(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)))}|t_1\right)$ and $\sigma^2(t_1)=\mathbb{E}(w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1,t_1)|t_1)$ thus

$$
\begin{aligned}
h_nA_1 &= \frac{h_n}{n}\mathbb{E}\left(a^2(t_1)b_H(t_1)\sigma^2(t_2)\sigma^2(t_3)K_{h_n}^2(t_2-t_1)K_{h_n}^2(t_3-t_1)\right) \\
&= \frac{h_n}{nh_n^2}\int a^2(t_1)b_H(t_1)\sigma^2(t_1+uh_n)\sigma^2(vh_n+t_1)K^2(u)K^2(v)du\,dv\,dt_1=\frac{1}{nh_n}O(1)
\end{aligned}
$$

then, we have that $h_nA_1\to 0$. On the other hand, using (P.13), we get the following bound

$$
\begin{aligned}
|A_2| &\le \frac{2}{nh_n^4}\mathbb{E}\left(a^2(t_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1))^4}{V^2(H(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)))}K^2((t_2-t_1)/h_n)K^2((t_3-t_1)/h_n)h_n\right) \\
&\le \frac{2}{nh_n^3}\int a^2(t_1)b(t_1)K^2((t_2-t_1)/h_n)K^2((t_3-t_1)/h_n)f(t_1)f(t_2)f(t_3)dt_1dt_2dt_3 \\
&\le \frac{2}{nh_n}\int a^2(t_1)b(t_1)K^2(u)K^2(v)f(t_1)f(uh_n+t_1)f(vh_n+t_1)dt_1\,du\,dv=\frac{1}{nh_n}O(1)
\end{aligned}
$$

which implies that $h_nA_2\to 0$. Finally, straighforward calculations lead to

$$
\begin{aligned}
h_nA_3 &= 2h_n\mathbb{E}\left(a(t_1)a(t_2)b(t_1)b(t_2)c(t_1,t_2,t_3)c(t_1,t_2,t_4)K_{h_n}(t_3-t_1)K_{h_n}(t_4-t_1)K_{h_n}(t_3-t_2)K_{h_n}(t_4-t_2)\right) \\
&= \frac{2}{h_n^3}\int a(t_1)a(t_2)b(t_1)b(t_2)c(t_1,t_2,t_3)c(t_1,t_2,t_4)K((t_3-t_1)/h_n) \\
&\quad K((t_4-t_1)/h_n)K((t_3-t_2)/h_n)K((t_4-t_2)/h_n)f(t_1)f(t_2)f(t_3)f(t_4)dt_1\,dt_2\,dt_3\,dt_4 \\
&= \frac{2}{h_n}\int a(t_1)a(t_2)b(t_1)b(t_2)c(t_1,t_2,uh_n+t_1)c(t_1,t_2,vh_n+t_1)K(u)K(v)K((uh_n+t_1-t_2)/h_n) \\
&\quad K((v+h_n+t_1-t_2)/h_n)f(t_1)f(t_2)f(uh_n+t_1)f(uh_n+t_1)dt_1\,dt_2\,du\,dv \\
&= \frac{2}{h_n}\int a(h_nz+t_2)a(t_2)b(h_nz+t_2)b(t_2)c(h_nz+t_2,t_2,uh_n+h_nz+t_2)c(h_nz+t_2,t_2,vh_n+h_nz+t_2) \\
&\times\quad K(u)K(v)K((uh_n+h_nz+t_2)/h_n)K((v+h_n+h_nz+t_2)/h_n)f(h_nz+t_2) \\
&\times\quad f(t_2)f(uh_n+t_1)f(uh_n+t_1)dz\,dt_2\,du\,dv
\end{aligned}
$$

and so, h_nA_3 converges to $2\int a^2(t_2)b^2(t_2)c^2(t_2,t_2,t_2)K(u)K(v)K(u+z)K(v+z)f^4(t_2)dt_2du\,dv\,dz=$ $2\,\mathbb{E}(a^2(t)b^2(t)c^2(t,t,t)f^3(t))\int[K*K(u)]^2du$.

Using that $c^2(t_2,t_2,t_2)=\mathbb{E}(\mathbb{E}(V_{2,2}^2|(\mathbf{x}_2,t_2))|t_2)=\mathbb{E}(w_1^2(\mathbf{x}_2)\sigma^2(\mathbf{x}_2,t_2)|t_2)$, we get that

$$
\begin{aligned}
&\mathbb{E}(a^2(t_2)b^2(t_2)c^2(t_2,t_2,t_2)f^3(t_2))=\mathbb{E}(a^2(t_2)b^2(t_2)\mathbb{E}(w_1^2(\mathbf{x}_2)\sigma^2(\mathbf{x}_2,t_2)|t_2)f^3(t_2)) \\
&=\mathbb{E}\left(\frac{w^2(t_1)}{v_0^4(t_1)f(t_1)}\left[\mathbb{E}\left(w_2(\mathbf{x}_1)\frac{H'(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1))}{V(H(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}_0+\eta_0(t_1)))}|t_1\right)\right]^2\mathbb{E}(w_1^2(\mathbf{x}_1)\sigma^2(\mathbf{x}_1,t_1)|t_1)\right)=c_{2,\Psi}\ ,
\end{aligned}
$$

concluding the proof. □

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